1 Genome sequencing and assembly

1.1 Extraction of genome DNA

Genomic DNA was extracted with the STE method. The harvested DNA was detected by the agarose gel electrophoresis and quantified by Qubit® 2.0 Fluorometer (Thermo Scientific).

1.2 Library construction

（1） Nanopore PromethION platform

Libraries for Nanopore sequencing was constructed with an insert size of 10 kb.

First, large fragments of DNA were recovered by Blue Pippin automatic nucleic acid fragment recovery system, and then repaired. Barcode was added by PCR-free method of EXP-NBD104 kit of Oxford Nanopore Technologies Company. Afterwards, the size of fragments was detected by AATI automatic capillary electrophoresis instrument, and the samples were isomolar mixed. Finally, the SQK-LSK109 connection kit was used to link the adapter, and the 10K library was constructed.

（2） Illumina NovaSeq platform

A total amount of 1μg DNA per sample was used as input material for the DNA sample preparations. Sequencing libraries were generated using NEBNext® Ultra™ DNA Library Prep Kit for Illumina (NEB, USA) following manufacturer’s recommendations and index codes were added to attribute sequences to each sample.

Briefly, the DNA sample was fragmented by sonication to a size of 350bp, then DNA fragments were end-polished, A-tailed, and ligated with the full-length adaptor for Illumina sequencing with further PCR amplification. At last, PCR products were purified (AMPure XP system) and libraries were analysed for size distribution by Agilent2100 Bioanalyzer and quantified using real-time PCR.

1.3 Sequencing

The whole genome of XXX was sequenced using Nanopore PromethION platform and Illumina NovaSeq PE150 at the Beijing Novogene Bioinformatics Technology Co., Ltd.

1.4 Assembly

Use **Unicycler** to combine PE150 data and Nanopore data to assemble, then compare the reads to the assembled sequence, count the distribution of sequencing depth, and distinguish whether the assembled sequence is a chromosomal sequence or a plasmid sequence according to sequence length and alignment, and check whether it is a circular genome.

(1) Preliminary assembly: First use SPAdes to assemble a frame diagram of the second-generation data;

(2) Create a bridge: Then use miniasm and Racon to add three generations

of data to the frame map, and use three generations of long-read data to create a bridge;

(3) Bridge application: When conflict occurs, Unicycler chooses a high score bridge according to the quality score corresponding to the created bridge;

(4) Judgement and final confirmation of loop formation:

1. Whether there is overlap at both ends of cotig;

2. Default dnaA or repA, start site correction and final confirmation through start_genes, screening chromosome and plasmid sequence Assembled into a circular genome (if it is a linear genome, it is a linear genome sequence), that is, the final

0gap complete sequence.

2 Genome Component prediction

Genome component prediction included the prediction of the coding gene, repetitive sequences, non-coding RNA, genomics islands, transposon, prophage, and clustered regularly interspaced short palindromic repeat sequences (CRISPR). The available steps were proceeded as follows:

1) For bacteria, we used the GeneMarkSprogram to retrieve the related coding gene.

2) The interspersed repetitive sequences were predicted using the RepeatMasker (http://www.repeatmasker.org/). The tandem Repeats were analyzed by the TRF (Tandem repeats finder).

3) Transfer RNA(tRNA) genes were predicted by the tRNAscan-SE . Ribosome RNA (rRNA) genes were analyzed by the rRNAmmer .Small nuclear RNAs(snRNA)were predicted by BLAST against the Rfam database.

4) The IslandPath-DIOMB

program was used to predict the Genomics Islands, and transposon PSI was used to predict the transposons based on the homologous blast method. The PHAST was used for the prophage

prediction(http://phast.wishartlab.com/) and the CRISPRFinder was used for the CRISPR identification.

3 Gene function

We used seven databases to predict gene functions. They were respective GO (Gene Ontology), KEGG (Kyoto Encyclopedia of Genes and Genomes), COG (Clusters of Orthologous Groups), NR (Non-Redundant Protein Database databases), TCDB (Transporter Classification Database), and, Swiss-Prot . A whole genome Blast search (Evalue less than 1e-5, minimal alignment length percentage larger than 40%) was performed against above seven databases. The secretory proteins were predicted by the Signal P database, and the prediction of Type I-VII proteins secreted by the pathogenic bacteria were based on the EffectiveT3 software. Meanwhile, we analyzed the secondary metabolism gene clusters by the antiSMASH .For pathogenic bacteria, we added the pathogenicity and drug resistance analyses. We used the PHI (Pathogen Host Interactions), VFDB (Virulence Factors of Pathogenic Bacteria), ARDB (Antibiotic Resistance Genes Database) to perform the above analyses.Carbohydrate-Active enzymes were predicted by the Carbohydrate-Active enZYmes Database .

4 Comparative genomics analysis

Comparative genomic analysis included the genomic synteny, the core genes and specific genes, gene family phylogenetic tree, SNP(Single Nucleotide Polymorphism), indel (insertion and deletion) and SV(Structural Variation) annotation, and genome visualization.

1)Genomic alignment between the sample genome and reference genome (or among more than two sample genomes) were performed using the MUMmer and LASTZ tools. Genomic synteny was analyzed based on the alignment results.

2) Core genes and specific genes were analyzed by the CD-HIT rapid clustering of similar proteins software with a threshold of 50% pairwise identity and 0.7 length difference cutoff in amino acid. Then the venn figure was drawn to show their relationships among the samples. 3) Gene family was constructed by the gene of XXX and XXX, using the multiple softwares: Blast was used to pairwise align all genes and eliminate the redundancy by solar and carried out gene family clustering treatment based on the alignment results with Hcluster_sg software.

4) The phylogenetic tree was constructed by the TreeBeST or PhyML and the setting of bootstraps was 1,000 with the orthologous genes.

5) SNP, indel and SV were found by the genomic alignment results among samples by the MUMmer and LASTZ we mentioned them before.

5 Genome overview was created by Circos to show the annotation information.